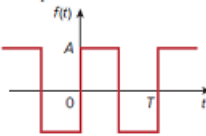
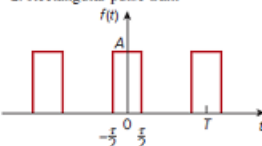
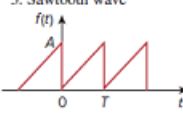
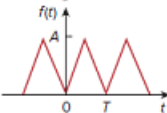
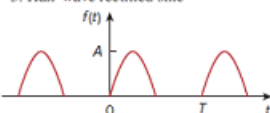
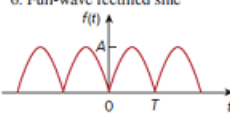


Problem

Write a program to compute the Fourier coefficients (up to the 10th harmonic) of the square wave in Table 17.3 with $A = 10$ and $T = 2$.

Table 17.3: The Fourier series of common functions.

Function	Fourier series
1. Square wave 	$f(t) = \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\omega_0 t$
2. Rectangular pulse train 	$f(t) = \frac{A\tau}{T} + \frac{2A}{T} \sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{n\pi\tau}{T} \cos n\omega_0 t$
3. Sawtooth wave 	$f(t) = \frac{A}{2} - \frac{A}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\omega_0 t}{n}$
4. Triangular wave 	$f(t) = \frac{A}{2} - \frac{4A}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\omega_0 t$
5. Half-wave rectified sine 	$f(t) = \frac{A}{\pi} + \frac{A}{2} \sin \omega_0 t - \frac{2A}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos 2n\omega_0 t$
6. Full-wave rectified sine 	$f(t) = \frac{2A}{\pi} - \frac{4A}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos n\omega_0 t$

Step-by-step solution

Step 1 of 1

From table 17.3, it is evident that $a_n = 0$,

$$b_n = \frac{4A}{\pi[2n-1]}, \quad A = 10$$

A FORTRAN program to calculate b_n is shown below the result is also shown.

C for problem 17.79

Dimension B (20)

A=10

PIE = 3.142

C = 4.* A/PIE

DO 10 N=1, 10

B (N) = C/ (2.*FLOAT (N)-1.)

PRINT *, N, B (N)

10 CONTINUE

STOP

$n \quad b_n$

1 12.731

2 4.243

3 2.546

4 1.8187

5 1.414

6 1.1573

7 0.9793

8 0.8487

9 0.7498

10 0.67